

Tecnológico Nacional de México

Campus Saltillo

Ingeniería en Sistemas Computacionales

Computo en la nube

5.8 Instalación del entorno LAMP en una máquina virtual en infraestructura de la nube.

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1. Actualizar sistema

```
hugo@MV-Hugo:~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1919 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [346 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:16 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [17.1 kB]

Running kernel seems to be up-to-date.

Restarting services...
/etc/needrestart/restart.d/systemd-manager
systemctl restart chrony.service multipathd.service polkit.service rsyslog.service ssh.service systemd-journald.service
systemd-networkd.service systemd-resolved.service systemd-udev.service udisks2.service walinuxagent.service

Service restarts being deferred:
systemctl restart ModemManager.service
/etc/needrestart/restart.d/dbus.service
systemctl restart networkd-dispatcher.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
hugo @ session #1: apt[1527], sshd[1053]
hugo @ user manager service: systemd[1058]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
hugo@MV-Hugo:~$
```



2. Instalar Apache

```
no VM guests are running outdated hypervisor (qemu) binaries on this host.
hugo@MV-Hugo:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64
  liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 0 not upgraded.
Need to get 2091 kB of archives.
After this operation, 8113 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [11.2 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9116 B]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-3build2 [166 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.11 [1335 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.11 [163 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.11 [98.2 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu8.11 [90.2 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Fetched 2091 kB in 0s (50.3 MB/s)
Preparing packages...
```

```
hugo@MV-Hugo:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-29 01:30:37 UTC; 24s ago
     Docs: https://httpd.apache.org/docs/2.4/
  Process: 709 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
 Main PID: 875 (apache2)
    Tasks: 6 (limit: 975)
   Memory: 13.8M (peak: 33.7M)
      CPU: 93ms
   CGroup: /system.slice/apache2.service
           └─875 /usr/sbin/apache2 -k start
             └─907 /usr/sbin/apache2 -k start
               └─908 /usr/sbin/apache2 -k start
                 └─909 /usr/sbin/apache2 -k start
                   └─910 /usr/sbin/apache2 -k start
                     └─911 /usr/sbin/apache2 -k start

Apr 29 01:30:36 MV-Hugo systemd[1]: Starting apache2.service - The Apache HTTP Server...
Apr 29 01:30:37 MV-Hugo apachectl[761]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, please see the README file for the appropriate configuration.
Apr 29 01:30:37 MV-Hugo systemd[1]: Started apache2.service - The Apache HTTP Server.
```



3. Instalar PHP

```
hugo@MV-Hugo:~$ sudo apt install php libapache2-mod-php -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libapache2-mod-php8.3 php-common php8.3 php8.3-cli php8.3-common php8.3-opcache php8.3-readline
Suggested packages:
  php-pear
The following NEW packages will be installed:
  libapache2-mod-php libapache2-mod-php8.3 php php-common php8.3 php8.3-cli php8.3-common php8.3-opcache
  php8.3-readline
0 upgraded, 9 newly installed, 0 to remove and 0 not upgraded.
Need to get 4923 kB of archives.
After this operation, 22.4 MB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 php-common all 2:93ubuntu2 [13.9 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-common amd64 8.3.6-0ubuntu0.24.04.8 [740 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-opcache amd64 8.3.6-0ubuntu0.24.04.8 [371 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-readline amd64 8.3.6-0ubuntu0.24.04.8 [13.5
hugo@MV-Hugo:~$ php -v
PHP 8.3.6 (cli) (built: Mar 20 2026 02:32:55) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.3.6, Copyright (c) Zend Technologies
    with Zend OPcache v8.3.6, Copyright (c), by Zend Technologies
hugo@MV-Hugo:~$
```

4. Instalar MySQL

```
hugo@MV-Hugo:~$ sudo apt install mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin
  libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl
  libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64 libtimedate-perl
  liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common
  mysql-server-8.0 mysql-server-core-8.0
Suggested packages:
  libdata-dump-perl libipc-sharedcache-perl libio-compress-brotli-perl libbusiness-isbn-perl libregexp-ipv6-perl
  libwww-perl mailx tinycat
The following NEW packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin
  libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl
  libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64 libtimedate-perl
  liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common
  mysql-server mysql-server-8.0 mysql-server-core-8.0
0 upgraded, 28 newly installed, 0 to remove and 0 not upgraded.
Need to get 29.6 MB of archives.
After this operation, 243 MB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 mysql-common all 5.8+1.1.0build1 [6746 B]
```

```
hugo@MV-Hugo:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-29 01:30:37 UTC; 24s ago
     Docs: https://httpd.apache.org/docs/2.4/
  Process: 709 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
 Main PID: 875 (apache2)
    Tasks: 6 (limit: 975)
   Memory: 13.8M (peak: 33.7M)
      CPU: 93ms
   CGroup: /system.slice/apache2.service
           └─875 /usr/sbin/apache2 -k start
             └─907 /usr/sbin/apache2 -k start
               └─908 /usr/sbin/apache2 -k start
                 └─909 /usr/sbin/apache2 -k start
                   └─910 /usr/sbin/apache2 -k start
                     └─911 /usr/sbin/apache2 -k start

Apr 29 01:30:36 MV-Hugo systemd[1]: Starting apache2.service - The Apache HTTP Server...
Apr 29 01:30:37 MV-Hugo apachectl[761]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, please see the /usr/sbin/apachectl(8) man page.
Apr 29 01:30:37 MV-Hugo systemd[1]: Started apache2.service - The Apache HTTP Server.
```

5. Probar en el navegador

Not secure 20.220.163.71



Apache2 Default Page

Ubuntu

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/  
|-- apache2.conf  
|  
|   |-- ports.conf  
|-- mods-enabled  
|  
|   |-- *.load  
|   |-- *.conf  
|-- conf-enabled  
|  
|   |-- *.conf  
|-- sites-enabled  
|  
|   |-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- `ports.conf` is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
- Configuration files in the `mods-enabled/`, `conf-enabled/` and `sites-enabled/` directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual

6. Conclusiones

En esta práctica se realizó la configuración básica de una máquina virtual en Azure, donde se actualizó el sistema e instalaron los servicios Apache, MySQL y PHP. Se comprobó su funcionamiento mediante pruebas, logrando contar con un entorno listo para el desarrollo web. Esto permitió entender el proceso inicial para preparar un servidor en la nube.